Project Synopsis: Human Activity Recognition Using 2D Pose Estimation and LSTM

1. Project Title

Human Activity Recognition Using 2D Pose Estimation and LSTM Recurrent Neural Network

2. Introduction

Human Activity Recognition (HAR) is an important application of Artificial Intelligence and Computer Vision that enables computers to identify and classify human actions from video data. This project uses **2D body pose keypoints** extracted from video frames and a **Stacked Long Short-Term Memory (LSTM) Recurrent Neural Network** to recognize six different human activities. Instead of processing complete RGB images, the system uses lightweight pose coordinates, making it faster and computationally efficient while maintaining high accuracy.

3. Objective

The main objective of this project is to develop a deep learning-based system capable of recognizing human activities from 2D body poses. The system aims to:

- Detect and classify human activities accurately.
- Reduce computational complexity using 2D pose estimation.
- Improve activity recognition using LSTM networks.
- Demonstrate the application of deep learning in computer vision.

4. Problem Statement

Traditional activity recognition systems process complete video frames, requiring high computational resources and large datasets. This project addresses the problem by using only human body keypoints extracted through pose estimation, significantly reducing the amount of input data while achieving excellent recognition accuracy.

5. Technologies Used

Technology	Purpose
Python	Programming Language
TensorFlow 1.x	Deep Learning Framework
LSTM (RNN)	Activity Classification
OpenPose	Human Pose Estimation
Flask	Web Application
HTML, CSS	User Interface
NumPy	Data Processing
Matplotlib	Data Visualization
Google Colab	Model Training

6. System Modules

Pose Estimation Module

- Extracts 2D body keypoints from video frames.
- Converts body movements into numerical coordinates.

Data Processing Module

- Prepares pose data for model training.
- Creates time-sequence windows for activity recognition.

Deep Learning Module

- Uses a two-layer stacked LSTM network.
- Learns temporal movement patterns.
- Predicts the performed activity.

Web Application Module

- Provides an interface for uploading videos.
- Displays activity recognition results.

7. Activity Classes

The system recognizes the following six activities:

- Jumping
- Jumping Jacks
- Boxing
- Waving One Hand
- Waving Two Hands
- Clapping Hands

8. Working Process

1. User uploads or records a video.
2. OpenPose extracts body keypoints.
3. Pose coordinates are converted into feature vectors.
4. Sequential data is passed to the LSTM model.
5. The trained model predicts the activity.
6. The recognized activity is displayed on the web application.

9. Features

- Human activity recognition using deep learning.
- Uses lightweight 2D pose estimation.
- More than 90% prediction accuracy.
- Real-time activity classification.
- Flask-based web interface.
- Fast and efficient processing.

10. Advantages

- Lower computational cost than RGB-based methods.
- High recognition accuracy.
- Suitable for real-time applications.
- Easy deployment on standard cameras.
- Scalable for future enhancements.
- User-friendly interface.

11. Future Enhancements

- Support additional human activities.
- Real-time webcam-based recognition.
- Mobile application development.
- TensorFlow 2.x implementation.

- Cloud deployment.
- Integration with healthcare, fitness, and surveillance systems.

12. Applications

- Smart Surveillance Systems
- Healthcare Monitoring
- Sports Performance Analysis
- Fitness Tracking
- Human-Computer Interaction
- Robotics
- Elderly Care Systems
- Security Monitoring

13. Conclusion

The **Human Activity Recognition Using 2D Pose Estimation and LSTM** project demonstrates an efficient approach for recognizing human activities through deep learning. By combining OpenPose for pose estimation with a stacked LSTM neural network, the system accurately classifies human movements while reducing computational requirements. With an accuracy of over **90%**, this project highlights the effectiveness of sequence-based deep learning models in computer vision and serves as a strong foundation for future real-time activity recognition applications.